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Motor, blower, and refrigerator

Abstract

A motor includes: a bobbin having an insertion hole; a coil wound around the bobbin; a stator core having an excitation part that is inserted into the insertion hole; a rotor that is disposed rotatably with respect to the stator core; and a resin mold covering the bobbin, the coil, and the stator core. On a first opening end of the insertion hole, an exposed portion having an exposed surface that is exposed from the resin mold is formed. The exposed portion is positioned between the stator core and the resin mold.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national stage application of the PCT International Application No. PCT/JP2018/026438 filed on Jul. 13, 2018, which claims the benefit of foreign priority of Japanese patent application 2017-145991 filed on Jul. 28, 2017, the contents all of which are incorporated herein by reference.

TECHNICAL FIELD

(2) This disclosure relates to a motor, a blower, and a refrigerator. Particularly, this disclosure relates to a motor for driving an impeller of a blower which is mounted in a refrigerator.

BACKGROUND ART

(3) The motor is used in various kinds of electric equipment. For example, the motor is used in a blower mounted in a refrigerator or the like. The motor used in the blower includes a stator, and a rotor which is rotated by the stator, for example. The rotor has a shaft which forms a center of rotation of the rotor. The blower further includes an impeller which is mounted on the shaft of the rotor, and supplies air by rotating the rotor using the stator.

(4) FIG. 19 is a cross-sectional view of conventional blower 100 disclosed in PTL 1.

(5) FIG. 19 shows conventional blower 100 disclosed in PTL 1. In conventional blower 100, a motor includes rotor 101, stator core 102 for rotating rotor 101, coil 104 which is wound around teeth portions (excitation parts) of stator core 102 by way of bobbin 103, and circuit board 105 on which electronic parts for controlling the supply of electricity to coil 104 is mounted.

(6) Bobbin 103, coil 104, circuit board 105, and a portion of stator core 102 disposed around bobbin 103, coil 104, and circuit board 105 are covered by resin mold 106. Impeller 107 is

mounted on shaft **108** of rotor **101**. In blower **100**, impeller **107** is rotated in an interlocking manner with the rotation of rotor **101**, so that air is sucked into a casing as indicated by an arrow in FIG. **19**. Air sucked into the casing is discharged from the casing through a discharge port formed on a left side of the casing in the drawing. With such a configuration, blower **100** can supply air. (7) However, in the configuration of the conventional motor, there is a possibility that water intrudes into the motor from a boundary between the stator core and the resin mold, and the water reaches the coil which is covered by the resin mold thus giving rise to a concern that a layer short of the coil occurs.

CITATION LIST

Patent Literature

(8) PTL 1: Unexamined Japanese Patent Publication No. H11-148484

SUMMARY

(9) This disclosure is made to overcome the above-mentioned drawback. It is an object of the present disclosure to provide a motor or the like which can suppress the occurrence of a layer short of a coil which is covered by a resin mold.

(10) To achieve the above-mentioned object, according to an aspect of this disclosure, a motor includes: a bobbin having an insertion hole; a coil wound around the bobbin; a stator core having an excitation part that is inserted into the insertion hole; a rotor disposed rotatably with respect to the stator core; and a resin mold covering the bobbin, the coil, and the stator core. On a first opening end of the insertion hole, an exposed portion having an exposed surface that is exposed from the resin mold is formed. The exposed portion is positioned between the stator core and the resin mold.

(11) According to another aspect of the present disclosure, a blower includes: a motor; and an impeller mounted on a shaft of a rotor.

(12) According to another aspect of the present disclosure, a refrigerator includes: a blower; a control unit configured to control the blower; and a refrigerating compartment or a freezing compartment, wherein the blower is configured to supply cool air to the refrigerating compartment or the freezing compartment.

(13) With the configuration of the present disclosure, the occurrence of a layer short of a coil covered by a resin mold can be suppressed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is an external perspective view of a motor according to a first exemplary embodiment.

(2) FIG. **2** is a front view of the motor according to the first exemplary embodiment.

(3) FIG. **3** is a side view of the motor according to the first exemplary embodiment.

(4) FIG. **4** is an exploded perspective view of the motor according to the first exemplary embodiment.

(5) FIG. **5** is a view showing a state where a resin mold of the motor in region V surrounded by a broken line in FIG. **4** is omitted.

(6) FIG. **6** is a view showing a state where a stator core in FIG. **5** is disassembled into a first segment core and a second segment core.

(7) FIG. **7** is an enlarged cross-sectional perspective view showing a main part of the motor according to the first exemplary embodiment.

(8) FIG. **8** is an enlarged cross-sectional view showing the main part of the motor according to the first exemplary embodiment.

(9) FIG. **9** is a schematic view showing one example of a refrigerator in which a blower using the motor according to the first exemplary embodiment is mounted.

- (10) FIG. 10 is a view showing a state where water droplet adheres to a surface of the stator core of the motor according to the first exemplary embodiment.
- (11) FIG. 11 is an enlarged cross-sectional view showing a main part of a motor according to a comparison example.
- (12) FIG. 12 is an enlarged cross-sectional view showing a main part of the motor in region XII surrounded by a broken line in FIG. 8.
- (13) FIG. 13 is an enlarged cross-sectional view showing a main part of a motor according to a second exemplary embodiment.
- (14) FIG. 14 is an enlarged cross-sectional view showing a main part of a motor according to a third exemplary embodiment.
- (15) FIG. 15 is a side cross-sectional view of the motor according to the third exemplary embodiment.
- (16) FIG. 16 is another side cross-sectional view of the motor according to the third exemplary embodiment.
- (17) FIG. 17 is an enlarged cross-sectional view showing a main part of a motor according to a modification.
- (18) FIG. 18 is an enlarged cross-sectional view showing a main part of a motor according to another modification.
- (19) FIG. 19 is a cross-sectional view of a conventional blower disclosed in PTL 1.

DESCRIPTION OF PREFERRED EMBODIMENTS

- (20) Exemplary embodiments of the present disclosure are described with reference to the drawings hereinafter. The exemplary embodiments described hereinafter show preferred specific examples of the present disclosure. Therefore, numeric values, shapes, materials, constitutional elements, arrangement positions and connection modes of the constitutional elements and the like which shown in the following exemplary embodiments are merely examples, and are not intended to limit the present disclosure. Further, out of the constitutional elements in following exemplary embodiments, the constitutional elements which are not described in independent claims describing an uppermost concept of the present disclosure are described as optional constitutional elements.
- (21) The drawings are schematic views and are not always exactly illustrated. In the drawings, substantially identical constitutional elements are given the same symbols, and the repeated description is omitted or simplified as appropriate.

First Exemplary Embodiment

- (22) Motor 1 according to a first exemplary embodiment is described with reference to FIG. 1 to FIG. 6. FIG. 1 is an external perspective view of motor 1 according to the first exemplary embodiment. FIG. 2 is a front view of motor 1 according to the first exemplary embodiment. FIG. 3 is a side view of motor 1 according to the first exemplary embodiment. FIG. 4 is an exploded perspective view of motor 1 according to the first exemplary embodiment. FIG. 5 is a view showing a state where resin mold 60 of motor 1 in region V surrounded by a broken line in FIG. 4 is omitted. FIG. 6 is a view showing a state where stator core 10 in FIG. 5 is disassembled into first segment core 11 and second segment core 12. In FIG. 5 and FIG. 6, coil 40 is shown as an integral body. In FIG. 6, circuit board 50 is shown by a broken line.
- (23) As shown in FIG. 1 to FIG. 6, motor 1 according to this exemplary embodiment includes: stator core 10; rotor 20 which is disposed rotatably with respect to stator core 10; bobbin 30 having insertion hole 31; coil 40 wound around bobbin 30; circuit board 50; resin mold 60; and bearing unit 70.
- (24) Stator core 10 is an iron core which forms a stator for generating a magnetic force for rotating rotor 20. In this exemplary embodiment, stator core 10 includes: first segment core 11; and second segment core 12 which is connected to first segment core 11. First segment core 11 and second segment core 12 are each formed of a laminated body formed by laminating a plurality of (for example, twelve) electromagnetic steel plates. However, the present disclosure is not limited to

such a configuration, and first segment core **11** and second segment core **12** may be a bulk body formed of a magnetic material.

(25) First segment core **11** is formed so as to surround rotor **20**. To be more specific, as shown in FIG. 5 and FIG. 6, first segment core **11** has fitting hole **11a** into which rotor **20** is inserted by fitting. A shape of fitting hole **11a** as viewed in a plan view is substantially a circular shape. First segment core **11** may be formed of a pair of segment cores which are split in two in a lateral direction.

(26) First segment core **11** has legs lib. In this exemplary embodiment, first segment core **11** has two legs lib. Two legs lib are formed in a downwardly protruding manner from both ends of first segment core **11** in a width direction in the drawing (FIG. 6).

(27) Second segment core **12** has recesses **12a** which are connected to distal ends (connecting portions) of legs **11b** of first segment core **11**. In this exemplary embodiment, two recesses **12a** are provided at positions corresponding to legs lib of first segment core **11**. Recesses **12a** of second segment core **12** and distal ends of legs lib of first segment core **11** have shapes which allow fitting engagement between recesses **12a** and the distal ends. That is, recesses **12a** and the distal ends have circular shapes as viewed in a plan view, for example.

(28) Second segment core **12** is an elongated I-shaped core. As described later, coil **40** is wound around second segment core **12** by way of bobbin **30**. Second segment core **12** is inserted into insertion hole **31** of bobbin **30** as an excitation part of stator core **10**.

(29) Rotor **20** is a rotor which rotates due to a magnetic force of stator core **10**, and is surrounded by stator core **10**. Rotor **20** is disposed in fitting hole **11a** of first segment core **11** of stator core **10** rotatably. A minute air gap exists between an outer surface of rotor **20** and an inner surface of fitting hole **11a** of stator core **10**.

(30) Rotor **20** shown in FIG. 4 is formed of one set of an N pole and an S pole along a rotational direction of rotor **20**. As one example, rotor **20** is formed of; rotor body **21** which is formed by molding a plastic magnet material into a cylindrical shape; and shaft **22** which penetrates a center of rotor body **21**.

(31) Rotor **20** may be formed of a plurality of N poles and a plurality of S poles where an N pole and an S pole are alternately and repeatedly arranged along the rotational direction of rotor **20**. Rotor **20** is formed of rotor body **21** having a magnetic force, and shaft **22** which penetrates the center of rotor body **21**. Rotor body **21** is formed of; a rotor core (iron core) made of a magnetic material; and a plurality of permanent magnets fixed to the rotor core. The permanent magnets may be embedded in the rotor core, or may be mounted on a side surface of the rotor core. For example, rotor **20** may be formed such that a magnet insertion hole is formed in the rotor core, and the permanent magnet is embedded in the magnet insertion hole. The rotor core is formed of a laminated body formed by laminating a plurality of electromagnetic steel plates which are fixed to each other, for example. However, the present disclosure is not limited to such a configuration, and the rotor core may be a bulk body formed of a magnetic material. The permanent magnet is a bond magnet, a ceramic magnet, a metal magnet or the like, for example.

(32) Shaft **22** of rotor **20** is a rotary shaft of rotor **20**. That is, rotor **20** rotates using shaft **22** as a center of rotation. Shaft **22** is a rotation output shaft which rotates in an interlocking manner with the rotation of rotor body **21**. Shaft **22** is formed of a metal rod, for example.

(33) Rotor **20** is held by bearing units **70**. Bearing units **70** support one end of shaft **22** of rotor **20** rotatably by bearings or the like.

(34) As shown in FIG. 5 and FIG. 6, bobbin **30** is a stator-use bobbin mounted on stator core **10**. Insertion hole **31** is formed in bobbin **30**, and second segment core **12** which is an excitation part of stator core **10** is inserted into insertion hole **31**. That is, bobbin **30** is mounted on second segment core **12**.

(35) Bobbin **30** is a coil bobbin around which coil **40** is wound. Bobbin **30** is formed of an insulating material such as a resin material. In this exemplary embodiment, bobbin **30** is formed of

a thermoplastic resin, for example. However, a material for forming bobbin **30** is not limited to the thermoplastic resin provided that the material has an insulating property. For example, as the thermoplastic resin, poly butylene terephthalate, nylon, polyethylene terephthalate, an acrylonitrile butadiene styrene (ABS) resin or the like can be used.

(36) Bobbin **30** is a hollow member having insertion hole **31** which forms an internal space. Flanges **32** are formed on opening ends **36** of insertion hole **31** of bobbin **30**. In this exemplary embodiment, flange **32** is formed on each of opening ends **36** of insertion hole **31**. In the description made hereinafter, for facilitating the understanding of the description, assume one of opening ends **36** as first opening end **36a**, and assume the other of opening ends **36** positioned on a side opposite to first opening end **36a** as second opening end **36b**. As a matter of course, when the configuration adopted with respect to first opening end **36a** is applied to second opening end **36b**, substantially the same advantageous effect can be acquired. By forming flanges **32**, it is possible to suppress the occurrence of collapse of winding of coil **40** which is wound around a cylindrical portion of bobbin **30**.

(37) Although the detailed configuration is described later, bobbin **30** further includes exposed portion **33** having exposed surface **33a** which is exposed from resin mold **60**. Exposed portion **33** is formed on each of opening ends **36** of insertion hole **31** of bobbin **30**.

(38) Coil **40** is a winding coil which is wound around bobbin **30** mounted on stator core **10**. Coil **40** is a coil wire such as an enamel copper wire. By supplying electricity to coil **40**, a magnetic flux is generated in stator core **10** where second segment core **12** forms a start point. A material for forming a core wire portion of coil **40** is not limited to a material containing copper as a main component, and may be a material containing a metal material other than copper such as aluminum as a main component, or may be a material containing a conductive material other than a metal material as a main component.

(39) In this exemplary embodiment, a bobbin winding body is formed by winding single-phase coil **40** around bobbin **30**. By inserting second segment core **12** in the bobbin winding body, an excitation structural body is formed. A pair of lead terminals **34** is mounted on a portion of bobbin **30**. One end of coil **40** is connected to one of the pair of lead terminals **34**. The other end of coil **40** is connected to the other of the pair of lead terminals **34**.

(40) Circuit board **50** is a printed wiring substrate on which pattern wiring made of a conductive material such as copper is formed in a predetermined shape, for example. On circuit board **50**, an electronic part (not shown in the drawing) which controls the supply of electricity to coil **40** is mounted. One ends of lead terminals **34** of bobbin **30** and one ends of power supply wires **80** are connected to circuit board **50** by soldering, and connector **81** which holds power supply wires **80** is connected to circuit board **50**. Power supply wires **80** are lead wires, for example. The other ends of power supply wires **80** are connected to an external power source. The external power source supplies electricity to circuit board **50** via power supply wires **80**. External power source may be either an alternate current (AC) power source or a direct current (DC) power source.

(41) As shown in FIG. 1 to FIG. 4, resin mold **60** is a resin molded body formed in a predetermined shape. Resin mold **60** is formed using an insulation resin material. To be more specific, as a material for forming resin mold **60**, a thermosetting resin or a thermoplastic resin can be used. In this case, as the thermosetting resin, for example, an unsaturated-polyester-based resin, a phenol-based resin, an epoxy-based resin or the like can be used, for example. As the thermoplastic resin, for example, a polyethylene-terephthalate-based resin, a polybutylene-terephthalate-based resin or the like can be used.

(42) Resin mold **60** covers bobbin **30**, coil **40**, and a portion of stator core **10**. Resin mold **60** is fixed to stator core **10** in a state where resin mold **60** covers bobbin **30**, coil **40**, and the portion of stator core **10**. Resin mold **60** further covers circuit board **50**. By covering coil **40** and circuit board **50** by resin mold **60**, it is possible to protect coil **40** and circuit board **50**. Although coil **40** and circuit board **50** are completely covered by resin mold **60**, stator core **10** is partially covered by

resin mold **60**. Although bobbin **30** is substantially wholly covered by resin mold **60**, a portion of bobbin **30** is exposed from resin mold **60**.

(43) Hereinafter, the relationship between such bobbin **30** and resin mold **60** is described in detail with reference to FIG. **7** and FIG. **8** as well as FIG. **1** to FIG. **6**. FIG. **7** is an enlarged cross-sectional perspective view showing a main part of motor **1** according to the first exemplary embodiment. FIG. **8** is an enlarged cross-sectional view showing the main part of motor **1**. Cross-sectional portions in FIG. **7** and FIG. **8** are equal to each other.

(44) As shown in FIG. **7** and FIG. **8**, exposed portion **33** having exposed surface **33a** which is exposed from resin mold **60** is formed on opening ends **36** of insertion hole **31** of bobbin **30**. In this exemplary embodiment, exposed portion **33** is formed on each of opening ends **36a**, **36b** on both sides of insertion hole **31**.

(45) On each of opening ends **36** of insertion hole **31** of bobbin **30**, exposed portion **33** is positioned between stator core **10** and resin mold **60**. To be more specific, exposed portion **33** is positioned between second segment core **12** and resin mold **60**.

(46) Exposed portion **33** is formed as a protruding portion (protrusion) formed by protruding a portion of bobbin **30**. Exposed portion **33** protrudes outward from flange **32** along a direction of hole axis C of insertion hole **31** of bobbin **30**. To be more specific, exposed portion **33** has the structure where the whole circumference of an edge portion of an opening of insertion hole **31** of bobbin **30** is protruded in a rectangular-shaped frame body mounted on an outer surface of flange **32**.

(47) Exposed surface **33a** of exposed portion **33** is flush with a surface of portion **60a** of resin mold **60** that surrounds exposed portion **33**. Accordingly, exposed surface **33a** of exposed portion **33** is a picture-frame-like flat surface that surrounds second segment core **12** having a rectangular shape in cross section.

(48) In motor **1** having the above-mentioned configuration, when electricity is supplied to coil **40**, a field current flows in coil **40** and second segment core **12** (excitation part) is excited, so that a magnetic flux is generated. A magnetic flux generated in second segment core **12** is transmitted to first segment core **11**, so that a magnetic flux is generated on whole stator core **10**. A rotation magnetic field having two poles, that is, an S pole and an N pole is generated in stator core **10**. Rotor **20** rotates due to the rotation magnetic field of stator core **10**. That is, a magnetic force generated by an interaction between a magnetic flux of stator core **10** and a magnetic flux generated from the permanent magnet of rotor **20** generates a torque for rotating rotor **20** thus rotating rotor **20**.

(49) Motor **1** having the above-mentioned configuration can be used in a blower as a fan motor. In this case, a blower can be formed by housing motor **1** where an impeller (rotation blade) is mounted on a distal end of shaft **22** of rotor **20** in a casing.

(50) Such a blower can be used in electric equipment such as a refrigerator, an air conditioner or the like, for example. For example, in the refrigerator, the blower (indoor fan) is used for circulating cool air. The blower is used for cooling a compressor.

(51) Hereinafter, an application example of the blower which uses motor **1** is described with reference to FIG. **9**. FIG. **9** is a schematic view showing one example of refrigerator **2** on which blower **3** using motor **1** according to the first exemplary embodiment is mounted.

(52) As shown in FIG. **9**, refrigerator **2** includes blower **3** which uses motor **1**, controller **4** which controls blower **3**, and refrigerating compartment **5**. In refrigerator **2**, blower **3** supplies cool air which is cooled by a cooler to refrigerating compartment **5** by using impeller **6**. Controller **4** stops or starts the supply of air using blower **3** by controlling blower **3**. Controller **4** may adjust an amount of air supplied using blower **3** by controlling blower **3**. Particularly, controller **4** controls motor **1**. Cool air may be supplied to freezing compartment in place of refrigerating compartment **5**, or may be supplied to both refrigerating compartment **5** and the freezing compartment.

(53) As described above, refrigerator **2** of this exemplary embodiment includes blower **3**, controller

4 which controls blower 3, and refrigerating compartment 5 or the freezing compartment. Blower 3 supplies cool air to refrigerating compartment 5 or the freezing compartment.

(54) Next, the manner of operation and advantageous effects of motor 1 according to the first exemplary embodiment are described with reference to FIG. 10 to FIG. 12 by comparing motor 1 according to the first exemplary embodiment with a motor of a comparison example. FIG. 10 is a view showing a state where water droplet 90 adheres to a surface of stator core 10 of motor 1 according to the first exemplary embodiment. FIG. 11 is an enlarged cross-sectional view showing a main part of motor 1X of the comparison example. FIG. 12 is an enlarged cross-sectional view showing a main part of motor 1 according to the first exemplary embodiment in region XII surrounded by a broken line in FIG. 8.

(55) As shown in FIG. 10, motor 1 is disposed such that shaft 22 of rotor 20 is directed in a horizontal direction, for example. That is, a surface of stator core 10 is directed in a vertical direction.

(56) As shown in FIG. 10, there is a case where water droplet 90 adheres to the surface of stator core 10 of motor 1. For example, a blower for supplying cool air cooled by a cooler to a refrigerating compartment is used in a refrigerator. In the refrigerator, at the time of removing frost on the cooler, or when a door of the refrigerator is open, driving of the motor is stopped so as to stop the supply of air by the blower. When the supply of air by the blower is stopped, a temperature around motor 1 is increased so that condensation occurs and, as shown in FIG. 10, water droplet 90 is formed on a surface of stator core 10.

(57) Formed water droplet 90 slides and falls along the surface of stator core 10, and reaches a boundary between stator core 10 and resin mold 60. In this case, with respect to motor 1X according to the comparison example shown in FIG. 11, water intrudes into motor 1X through a boundary between stator core 10 and resin mold 60 as indicated by an arrow due to a capillary phenomenon. Further, there may be a case where intruded water advances along a boundary between flange 32 and resin mold 60 due to a capillary phenomenon, and reaches coil 40 which is covered by resin mold 60. As a result, the layer short of coil 40 occurs due to such water.

(58) In contrast, with respect to motor 1 according to this exemplary embodiment, as shown in FIG. 12, exposed portion 33 which is exposed from resin mold 60 is disposed so as to be positioned between stator core 10 and resin mold 60. Accordingly, between stator core 10 and resin mold 60, there is no boundary between bobbin 30 and resin mold 60.

(59) With such a configuration, even when water droplet 90 which adheres to the surface of stator core 10 slides and falls along the surface of stator core 10 and reaches a boundary between stator core 10 and bobbin 30, and intrudes into bobbin 30 due to a capillary phenomenon, as indicated by an arrow in FIG. 12, although water which intrudes into bobbin 30 intrudes into the motor along the boundary between stator core 10 and bobbin 30, the droplet does not reach coil 40.

(60) That is, due to the formation of exposed portion 33, even when water droplet 90 adheres to the surface of stator core 10, it is possible to avoid the intrusion of water into the motor through the boundary between bobbin 30 and resin mold 60. With such a configuration, it is possible to suppress reaching of water to coil 40 along the boundary between bobbin 30 and resin mold 60. Accordingly, it is possible to suppress the occurrence of a layer short of coil 40.

(61) As described above, motor 1 according to this exemplary embodiment includes: bobbin 30 having insertion hole 31; coil 40 wound around bobbin 30; stator core 10 having the excitation part which is inserted into insertion hole 31; rotor 20 disposed rotatably with respect to stator core 10; and resin mold 60 covering bobbin 30, coil 40 and stator core 10. On first opening end 36a of insertion hole 31, exposed portion 33 having exposed surface 33a which is exposed from resin mold 60 is formed. Exposed portion 33 is positioned between stator core 10 and resin mold 60.

(62) With such a configuration, between stator core 10 and resin mold 60, no boundary is formed between resin mold 60 and bobbin 30. With such a configuration, even when water intrudes into the motor through the boundary between members existing between stator core 10 and resin mold 60,

it is possible to suppress reaching of the water to coil **40**. Accordingly, it is possible to suppress the occurrence of a layer short of coil **40**. As a result, it is possible to realize highly reliable motor **1**.

(63) In this exemplary embodiment, bobbin **30** has flange **32** on first opening end **36a**, and exposed portion **33** protrudes outward from flange **32** along hole axis C direction of insertion hole **31** of bobbin **30**.

(64) With such a configuration, a water intrusion path where water reaches coil **40** from the surface of resin mold **60** or exposed surface **33a** can be elongated and hence, it is possible to further suppress reaching of water to coil **40**.

(65) In this exemplary embodiment, exposed surface **33a** is flush with a surface of portion **60a** of resin mold **60** that surrounds exposed portion **33**.

(66) With such a configuration, even when a water droplet slides and falls to an area around exposed portion **33** along the surface of stator core **10**, the water droplet does not stay on exposed portion **33** and easily passes. With such a configuration, it is possible to further suppress the intrusion of water into the motor through the boundary between the members existing between stator core **10** and resin mold **60**. Further, since exposed surface **33a** is flush with the surface of portion **60a** of resin mold **60** that surrounds exposed portion **33**, even when exposed portion **33** is formed on bobbin **30**, it is possible to make exposed portion **33** inconspicuous.

(67) In this exemplary embodiment, exposed surface **33a** has an annular shape.

(68) With such a configuration, exposed portion **33** can be formed so as to surround the whole circumference of first opening end **36a** of insertion hole **31** of bobbin **30** and hence, it is possible to further suppress the intrusion of water through the boundary between the members existing between stator core **10** and resin mold **60**.

(69) In this exemplary embodiment, the motor includes circuit board **50** on which the electronic part for controlling the supply of electricity to coil **40** is mounted, and circuit board **50** is covered by resin mold **60**.

(70) With such a configuration, not only coil **40** but also circuit board **50** can be protected by resin mold **60**. For example, by covering coil **40** and circuit board **50** by resin mold **60**, coil **40** and circuit board **50** can be insulated and protected. Accordingly, deterioration of coil **40** and circuit board **50** attributed to moisture and oxygen in air can be suppressed.

(71) In this exemplary embodiment, stator core **10** includes first segment core **11** and second segment core **12** connected to first segment core **11**. First segment core **11** is formed so as to surround rotor **20**. Second segment core **12** is the excitation part.

(72) With such a configuration, when second segment core **12** (excitation part) is excited so that a magnetic flux is generated, the magnetic flux is transferred to first segment core **11**, and a rotation magnetic field is generated in first segment core **11**. With such a configuration, rotor **20** which is disposed so as to be surrounded by first segment core **11** can be rotated.

Second Exemplary Embodiment

(73) Next, motor **1A** according to a second exemplary embodiment is described with reference to FIG. **13**. FIG. **13** is an enlarged cross-sectional view showing a main part of motor **1A** according to the second exemplary embodiment.

(74) In motor **1** of the first exemplary embodiment, exposed surface **33a** of exposed portion **33** of bobbin **30** has a uniform flat surface. On the other hand, in motor **1A** of this exemplary embodiment, as shown in FIG. **13**, exposed surface **33a** of exposed portion **33A** of bobbin **30A** includes a recessed surface.

(75) To be more specific, as shown in FIG. **13**, recess **33b** which is recessed from a surface of resin mold **60** is formed on exposed portion **33A**. In this exemplary embodiment, an inner surface of recess **33b** of exposed portion **33A** is formed in inclined surface **33c** which forms a portion of exposed surface **33a**. A distance between inclined surface **33c** of recess **33b** and a surface of stator core **10** decreases with distance from the surface of resin mold **60**. For example, a cross-sectional shape of a portion of recess **33b** which is recessed in a cutout shape is a right-angled triangle.

Recess **33b** can be formed by cutting out a corner portion of exposed portion **33** by chamfering the corner portion of exposed portion **33** in the first exemplary embodiment.

(76) As described above, also in motor **1A** of this exemplary embodiment, exposed portion **33A** having exposed surface **33a** exposed from resin mold **60** is formed on first opening end **36a** of insertion hole **31** of bobbin **30A**. Exposed portion **33A** is positioned between stator core **10** and resin mold **60**.

(77) With such a configuration, at a boundary between stator core **10** and resin mold **60**, a boundary between resin mold **60** and bobbin **30A** does not exist. Accordingly, even when water intrudes into the motor from the boundary between members existing between stator core **10** and resin mold **60**, it is possible to suppress reaching of water to coil **40**. Accordingly, also in this exemplary embodiment, it is possible to suppress the occurrence of a layer short of coil **40**. Accordingly, it is possible to realize highly reliable motor **1A**.

(78) In this exemplary embodiment, recess **33b** which is recessed from a surface of resin mold **60** is formed on exposed portion **33A**.

(79) With such a configuration, a water droplet around first opening end **36a** of insertion hole **31** of bobbin **30A** can easily intrude between bobbin **30A** and stator core **10** from recess **33b**. As a result, a water droplet minimally stays on a peripheral portion where the boundary between resin mold **60** and bobbin **30A** exists, that is, an area around first opening end **36a** of insertion hole **31** of bobbin **30A**. With such a configuration, it is possible to suppress intrusion of water into the motor through the boundary between resin mold **60** and bobbin **30A**. Accordingly, it is possible to further suppress the occurrence of a layer short of coil **40**. Accordingly, it is possible to realize more highly reliable motor **1A**.

(80) In this exemplary embodiment, an inner surface of recess **33b** is inclined surface **33c**, and the distance between inclined surface **33c** and the surface of stator core **10** decreases with distance from the surface of resin mold **60**.

(81) With such a configuration, a water droplet around first opening end **36a** of insertion hole **31** of bobbin **30A** can be easily introduced to recess **33b**. Accordingly, a water droplet can be easily drawn to an area between bobbin **30A** and stator core **10**. With such a configuration, it is possible to further suppress intrusion of water into the motor through the boundary between resin mold **60** and bobbin **30A**. Accordingly, it is possible to further suppress the occurrence of a layer short of coil **40**.

Third Exemplary Embodiment

(82) Motor **1B** according to a third exemplary embodiment is described with reference to FIG. **14** to FIG. **16**. FIG. **14** is an enlarged cross-sectional view showing a main part of motor **1B** according to the third exemplary embodiment. FIG. **15** is a side cross-sectional view of motor **1B**. FIG. **16** is another side cross-sectional view of motor **1B**.

(83) In motor **1** of the first exemplary embodiment, the surface of stator core **10** and the inner surface of insertion hole **31** of bobbin **30** are brought into close contact with each other. However, in motor **1B** of this exemplary embodiment, as shown in FIG. **14** to FIG. **16**, gap **G** (groove) is formed between a surface of stator core **10** and an inner surface of insertion hole **31** of bobbin **30B**. That is, gap **G** is formed between stator core **10** and bobbin **30B** as a space layer.

(84) As shown in FIG. **14**, gap **G** extends from first opening end **36a** of insertion hole **31** toward second opening end (**36b**) of insertion hole **31** disposed on a side opposite to first opening end **36a**. In this exemplary embodiment, gap **G** is formed over the whole length of insertion hole **31**.

(85) In this exemplary embodiment, as shown in FIG. **15**, protrusions **35** are formed on an inner side of bobbin **30B**. Four protrusions **35** are formed so as to be brought into contact with four (upper, lower, left, and right) surfaces of second segment core **12**. In this manner, by forming protrusions **35** on the inner side of bobbin **30B**, bobbin **30B** can be held on stator core **10**, particularly, on second segment core **12** while maintaining gap **G** between stator core **10** and bobbin **30B** in the circumferential direction.

(86) In this exemplary embodiment, as shown in FIG. 16, gaps G1 are formed on an inner side of bobbin 30E. Four gaps G1 are formed so as to bring into contact with four (upper, lower, left, and right) surfaces which second segment core 12 has. In this manner, by providing gaps G1 on the inner side of bobbin 30E, bobbin 30E can be held on stator core 10, particularly, on second segment core 12 while maintaining gaps G1 between stator core 10 and bobbin 30E in the circumferential direction.

(87) That is, as shown in FIG. 15, in a cross-sectional direction intersecting with hole axis C (cross-sectional direction shown in FIG. 15), an area where gap G and cross-sectional stator core 10 are brought into contact with each other, particularly, an area where gap G and second segment core 12 are brought into contact with each other is large. Accordingly, in this configuration, when a magnetic force is generated on stator core 10 so that stator core 10 is heated, water adhering to stator core 10 can be evaporated in an early stage.

(88) On the other hand, as shown in FIG. 16, in a cross-sectional direction intersecting with hole axis C (cross-sectional direction shown in FIG. 16), areas where gaps G1 and cross-sectional stator core 10 are brought into contact with each other, particularly, areas where gaps G1 and second segment core 12 are brought into contact with each other are small. With this configuration, water which is introduced into gaps G1 can pass through gaps G1 in an early stage due to a capillary phenomenon or the like.

(89) As described above, also in motor 1B of this exemplary embodiment, exposed portion 33 having exposed surface 33a exposed from resin mold 60 is formed on first opening end 36a of insertion hole 31 of bobbin 30B. Exposed portion 33 is positioned between stator core 10 and resin mold 60.

(90) With such a configuration, at a boundary between stator core 10 and resin mold 60, a boundary between resin mold 60 and bobbin 30B does not exist. Accordingly, even when water intrudes into the motor through the boundary between members existing between stator core 10 and resin mold 60, it is possible to suppress reaching of water to coil 40. Accordingly, also in this exemplary embodiment, it is possible to suppress the occurrence of a layer short of coil 40. As a result, it is possible to realize highly reliable motor 1B.

(91) In this exemplary embodiment, between the surface of stator core 10 and the inner surface of insertion hole 31, gap G, G1 which extends from first opening end 36a of opening ends 36 toward second opening end 36b of opening ends 36 positioned on a side of insertion hole 31 opposite to first opening end 36a is formed.

(92) With such a configuration, a water droplet around first opening end 36a of insertion hole 31 of bobbin 30B, 30E can easily intrude between bobbin 30B, 30E and stator core 10 through gaps G, G1. As a result, a water droplet minimally stays on a peripheral portion where the boundary between resin mold 60 and bobbin 30B, 30E exists, that is, an area around first opening end 36a of insertion hole 31 of bobbin 30B, 30E. With such a configuration, it is possible to suppress intrusion of water into the motor through the boundary between resin mold 60 and bobbin 30B, 30E.

Accordingly, it is possible to further suppress the occurrence of a layer short of coil 40.

Accordingly, it is possible to realize more highly reliable motor 1B. In this exemplary embodiment, gaps G, G1 are formed over the whole length of insertion hole 31.

(93) With such a configuration, it is possible to suppress water intruded into gaps G, G1 from staying in gaps G, G1. Accordingly, it is possible to realize more highly reliable motor 1B.

(94) (Modification)

(95) Although the motor, the blower, the refrigerator and the like according to this disclosure have been described heretofore with reference to the exemplary embodiments, the present disclosure is not limited to the above-mentioned exemplary embodiments.

(96) For example, in the first to third exemplary embodiments, exposed surface 33a of exposed portion 33 formed on bobbin 30 is flush with the surface of a portion of resin mold 60 surrounding the exposed portion. However, the present disclosure is not limited to such a configuration, and

exposed surface **33a** of exposed portion **33** may be positioned outside the surface of resin mold **60** as shown in FIG. **17**. Further, as shown in FIG. **18**, exposed surface **33a** may be positioned inside the surface of resin mold **60**. FIG. **17** is an enlarged cross-sectional view showing a main part of a motor according to a modification. FIG. **18** is an enlarged cross-sectional view showing a main part of a motor according to another modification. Bobbin **30C** of motor **1C** shown in FIG. **17** is formed such that exposed portion **33** protrudes from a surface of resin mold **60**. That is, in FIG. **17**, compared to the configuration shown in FIG. **12**, a protruding amount (height) of exposed portion **33** (protrusion) is large.

(97) Bobbin **30D** of motor **1D** shown in FIG. **18** is formed such that exposed portion **33** is recessed from the surface of resin mold **60**.

(98) In the first to third exemplary embodiments, exposed surface **33a** of exposed portion **33** is formed in an annular shape. However, the present disclosure is not limited to such a configuration. Provided that exposed portion **33** is formed such that at least a portion of bobbin **30** is exposed on the boundary between stator core **10** and bobbin **30**, it is possible to suppress the intrusion of a water droplet adhering to the surface of stator core **10** into bobbin **30**.

(99) The motors according to the first to third exemplary embodiments are used in the cooling blower. However, the application of the motors is not limited to the cooling blower, and the motors can be used in various electric equipment.

(100) In addition, the present disclosure includes modifications which those skilled in the art can obtain by adding changes to the first to third exemplary embodiments described above or modifications implemented by freely combining constitutional elements and functions described in the first to third exemplary embodiments without departing from the gist of the present disclosure.

(101) The technique of the present disclosure can be used in various electric equipment such as a motor, a blower, a refrigerator and the like. Particularly, the technique of the present disclosure is useful for a motor or the like which is used in an environment where a water droplet easily adheres to a surface of a stator core.

Claims

1. A motor comprising: a bobbin having an insertion hole having a first opening end and a second opening end positioned on a side opposite to the first opening end along an extending direction of the bobbin, a first flange disposed at the first opening end, a first exposed portion protruding from the first flange in the extending direction of the bobbin, a second flange disposed at the second opening end and a second exposed portion protruding from the second flange in the extending direction of the bobbin; a coil wound around the bobbin; a stator core having an excitation part that is inserted into the insertion hole; a rotor disposed rotatably with respect to the stator core; and a resin mold covering the bobbin, the coil, and the stator core, wherein: the resin mold covers the first flange and the second flange, the first exposed portion and the second exposed portion are formed around whole circumferences of an edge portion of the first opening of the first insertion hole and the second opening of the second insertion hole, respectively, the first exposed portion of the bobbin has a first exposed surface that is exposed from the resin mold formed on the first opening end and a second exposed surface that is exposed from the resin mold formed on the second opening end, the first exposed portion is positioned between the stator core and the resin mold, the first exposed surface being disposed partially around the first opening end of the insertion hole, and the second exposed portion is positioned between the stator core and the resin mold, the second exposed surface being disposed partially around the second opening end of the insertion hole.

2. The motor according to claim 1, wherein a recess that is recessed from a surface of the resin mold is formed in the first exposed portion.

3. The motor according to claim 2, wherein: the recess is an inclined surface, and the inclined

surface and a surface of the stator core forms a V-shape.

4. The motor according to claim 3, wherein a gap that extends from the first opening end toward a second opening end of the insertion hole opposite to the first opening end is formed between a surface of the stator core and an inner surface of the insertion hole.

5. The motor according to claim 4, wherein the gap is formed over a whole length of the insertion hole.

6. The motor according to claim 2, wherein a gap that extends from the first opening end toward a second opening end of the insertion hole opposite to the first opening end is formed between a surface of the stator core and an inner surface of the insertion hole.

7. The motor according to claim 6, wherein the gap is formed over a whole length of the insertion hole.

8. The motor according to claim 1, wherein a gap that extends from the first opening end toward a second opening end of the insertion hole opposite to the first opening end is formed between a surface of the stator core and an inner surface of the insertion hole.

9. The motor according to claim 8, wherein the gap is formed over a whole length of the insertion hole.

10. The motor according to claim 1, wherein the first exposed surface is flush with a surface of a portion of the resin mold, the portion partially surrounding and contacting a periphery of the first exposed portion.

11. The motor according to claim 1, further comprising a circuit board mounted with an electronic part for controlling a supply of electricity to the coil, and the circuit board is covered by the resin mold.

12. The motor according to claim 1, wherein the stator core includes: a first segment core; and a second segment core that is connected to the first segment core, the first segment core is formed so as to surround the rotor, and the second segment core is the excitation part.

13. A blower comprising: the motor according to claim 1; and an impeller mounted on a shaft of the rotor.

14. A refrigerator comprising: the blower according to claim 13; a control unit for controlling the blower; and a refrigerating compartment or a freezing compartment, wherein the blower is configured to supply cool air to the refrigerating compartment or the freezing compartment.

15. The motor according to claim 1, wherein: the first flange has a first surface contacting the coil and a second surface opposite to the first surface, the first exposed portion protrudes from the second surface of the first flange, and the resin mold covers the second surface of the first flange and a part of the first exposed portion.

16. The motor according to claim 15, wherein: the second flange has a first surface contacting the coil and a second surface opposite to the first surface, the second exposed portion protrudes from the second surface of the second flange, and the resin mold covers the second surface of the second flange and a part of the second exposed portion.

17. The motor according to claim 1, wherein: the first exposed portion is a rectangular-shaped frame body having four sides and the first opening end has a rectangular-shaped opening having four sides, and at least two sides of the rectangular-shaped frame body of the first exposed portion are disposed on corresponding two sides of the rectangular-shaped opening of the first opening end.
